

Section: Particles and Waves

Topic: Forces on Charged Particles

An electron enters a uniform electric field and is deflected.

(a)

State the direction of the electric field

✔ The electric field is directed **from the positive to the negative plate**, so the direction is **vertically downward** (as standard convention shows field lines go from + to −).

(b)

State the direction of the force acting on the electron

✔ The electron is **negatively charged**, so it experiences a force in the **opposite direction** to the electric field.
⇒ Force direction is upward.

(c)

Calculate the magnitude of the force on the electron

Use:

$$F = Eq$$

Given:

- $E = 3.0 \times 10^4 \text{ V m}^{-1}$
- $q = 1.6 \times 10^{-19} \text{ C}$

$$F = 3.0 \times 10^4 \times 1.6 \times 10^{-19} = 4.8 \times 10^{-15} \text{ N}$$

🧠 Revision Tips

- Electric field direction is **always from positive to negative**
- Force on a charge in an electric field:

$$F = Eq$$

If the charge is negative, the force acts **opposite** to the field

- Be careful with units:

$$E (\text{V m}^{-1}), \quad q (\text{C}), \quad F (\text{N})$$

- The force on a **single electron** is always tiny — use powers of 10 confidently!